NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL
Department of Information Technology

Ambulance Booking System

User Manual

For patients, ambulance crews and the control room

Team 16
IT303 — Software Engineering

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Document version	**1.0**
Date of issue	**16 October 2026**
Applies to	**Ambulance Booking v1.0.0**

This document describes an academic project. In a life-threatening emergency, telephone **108** directly.

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1 Introduction

1.1 Purpose of this manual

This manual serves the purpose of highlighting the different features which are available within the Ambulance Booking System and providing a detailed explanation of how each kind of user can carry them out. The system brings together three groups of people who, between them, complete a single ambulance journey: the **patient** who needs an ambulance, the **ambulance crew** who answer the request and drive it, and the **control room** who keep the service running and settle anything that goes wrong.

One application serves all three. The features a user is shown are decided by the account they sign in with, and not by a separate download, so this manual is organised in the same way: a common chapter on obtaining an account, and then one chapter for each role.

1.2 Who should read this

Reader	Read these chapters
Patients and families	Sections 2 and 3, and Section 7.
Drivers and operators	Sections 2 and 4, and Section 7.
Control room staff	Sections 2, 5 and 7.
Evaluators	All of it. Appendix B maps the specification onto the sections that demonstrate each requirement.

1.3 What a user needs before starting

- An Android phone with the **Ambulance Booking** application installed on it. The control room may instead be opened in any desktop browser.
- An email address that can be opened on the same device. The system sends a verification code to this address during registration, and the registration cannot be completed without it.
- A mobile number. The number is not used to sign in; it is the number the ambulance crew will call if they are unable to find the patient.
- A working internet connection. The application communicates with a server for every operation.

1.4 Abbreviations used in this document

Term	Meaning
EMG	Emergency request. Alerted to every nearby ambulance at once, irrespective of class.
BLS	Basic Life Support ambulance — oxygen, stretcher and a trained attendant.
ALS	Advanced Life Support ambulance — ventilator, defibrillator and a paramedic.
ICU	Mobile Intensive Care Unit — critical care with full monitoring.
PTV	Patient Transport Vehicle — a planned, non-emergency transfer.
OTP	One Time Password. A six-digit code, valid for five minutes, sent by email.
ETA	Estimated Time of Arrival.
Offer	A request presented to one ambulance, which expires after 45 seconds.

> **In a real emergency**
> This application was built as an academic project for the course IT303. In a life-threatening emergency, users should telephone **108** directly. A reminder to this effect is shown on every screen of the application where it can reasonably be placed.

2 Accessing the system

2.1 Choosing a role

On opening the application for the first time, users are presented with the role selection screen. Users need to select the role under which they are signing in.

- **Book an ambulance** — for patients and their families.
- **Drive an ambulance** — for drivers and ambulance operators.
- **Control room** — for administrators.

If an incorrect role is selected, users can return to this screen using the **Change mode** link at the top of the sign-in screen. Nothing is decided until the user signs in.

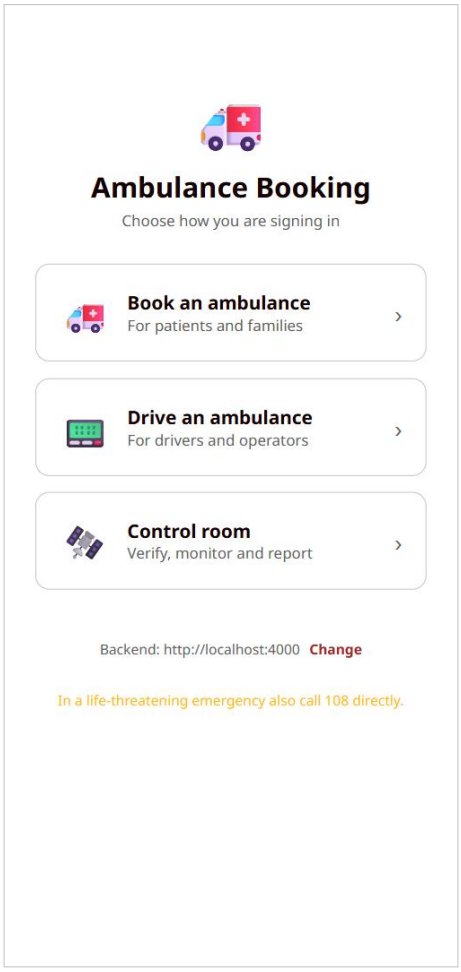


Figure 1. The role selection screen. The address of the server in use is shown at the bottom, and the reminder to call 108 is shown below it.

2.2 Registering a new account

Registration proves that the email address given belongs to the person registering. No account is created until the code sent to that address has been entered, so every account in the system has a working address behind it.

1. Open the application and select the appropriate role.
2. Click on **Register**.
3. Users need to fill in their full name, mobile number, email address and a password.
4. Click on **Email me a code**. No account has been created at this stage.
5. Users need to open their email and copy the six-digit code that has been sent to them. The code is valid for five minutes.
6. Enter the code in the application and click on **Verify and create account**.

The user is signed in as soon as the code is accepted. Drivers and operators are asked for their vehicle details immediately afterwards; this is described in Section 4.2.

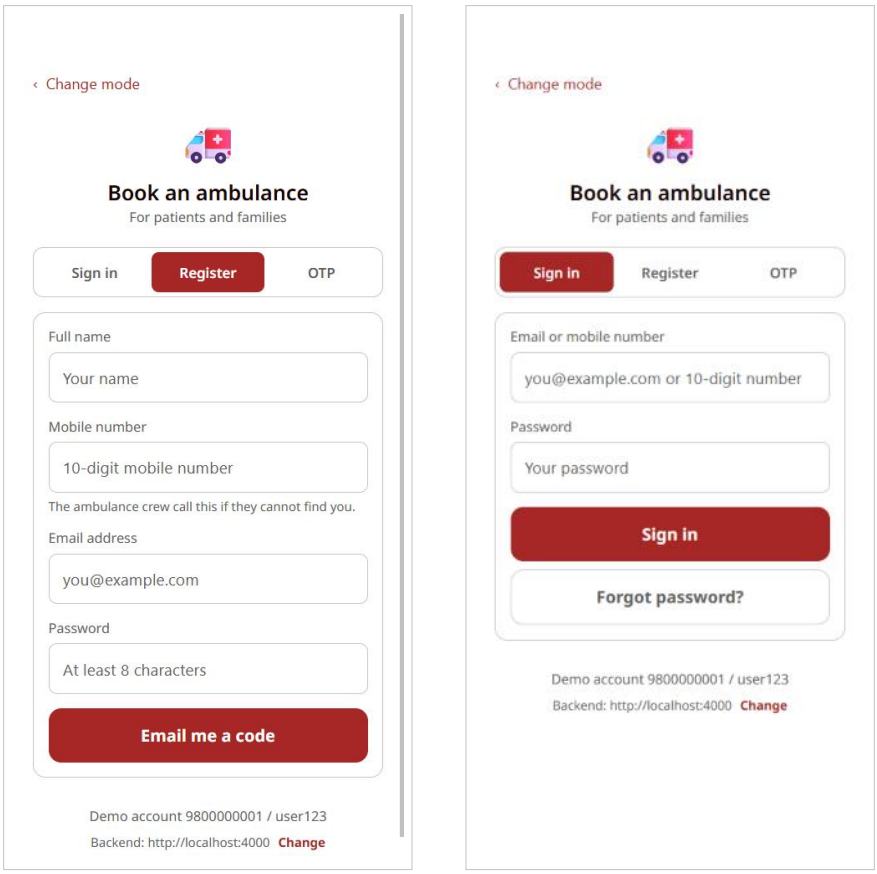


Figure 2. Left: the registration form, which asks for an email address. Right: the sign-in screen, which accepts either an email address or a mobile number.

2.2.1 Rules applied to the password

The application refuses a password that would be easy to guess, and reports the reason as the password is typed rather than after the form is submitted. A password is refused if it:

- is shorter than eight characters;
- appears on the list of most commonly used passwords;
- is a single character repeated, or a simple run such as `123456` or `abcdef`;
- contains the user's own name, mobile number or email address.

Either a long passphrase or a shorter mixture of letters, digits and symbols is accepted. The phrase `correct horse battery staple` is accepted; `Password1!` is not.

2.3 Signing in

Two methods of signing in are available.

- **With a password.** Users need to enter either their email address or their mobile number, followed by their password, and click on **Sign in**.
- **With a one time password.** Users need to click on **OTP**, enter their email address, and click on **Email me a code**. The code received by email is then entered in place of a password. This is useful when the password has been forgotten and the user does not wish to reset it.

2.4 Recovering a forgotten password

1. On the sign-in screen, click on **Forgot password?**
2. Enter the registered email address and click on **Email me a reset code**.
3. Enter the code received by email and choose a new password, observing the rules listed above.

> **Resetting a password ends every session**
> Once a password has been reset, every session on every device is signed out. This is deliberate: if the old password were known to somebody else, leaving their session open would defeat the purpose of the reset. Users need to sign in again with the new password.

3 The patient module

Patients have access to a certain set of actions that can be performed within the Ambulance Booking module for requesting and following an ambulance. They are illustrated below.

3.1 Actions available to the patient

- Request an ambulance to a pickup point, set in any of three ways:
  - by reading the device GPS;
  - by moving the map under a fixed centre pin;
  - by searching for an address or landmark.
- Select the class of ambulance required:
  - Emergency (EMG), Basic Life Support (BLS), Advanced Life Support (ALS), Mobile ICU (ICU) or Patient Transport (PTV).
- Follow the assigned ambulance on a live map, with an estimated time of arrival that is revised as the ambulance travels.
- Telephone the assigned driver directly from the application.
- Cancel a request at any time up to the point at which the patient is taken on board.
- View the itemised invoice and settle it by UPI, card, net banking or cash.
- Rate a completed trip from one to five stars, with a written comment.
- View every past trip under **Trips**.
- Report an ambulance crew for misconduct on a trip that has ended.

3.2 Module overview

The patient module presents four tabs along the bottom of the screen: **Book**, **Track**, **Trips** and **Account**.

3.3 Setting the pickup location

This is the single most important entry on the screen, and so three methods are offered. Users may use whichever is quickest in the circumstances.

- **Use my current location.** The application reads the device GPS. Where a fresh fix is slow to arrive, the last known position is used rather than waiting indefinitely.
- **Move the map.** The pin remains fixed at the centre of the map while the map itself moves beneath it. Users need to position the map so that the pin sits on the pickup point; the address shown below the map is revised as this is done.
- **Search for an address.** Users need to type a landmark, for example *NITK Surathkal main gate*, and select the correct entry from the results.

> **Confirm the address before requesting**
> The address shown beneath the pin is the address to which the ambulance will be sent. Users need to confirm that it reads as a place they recognise before placing the request.

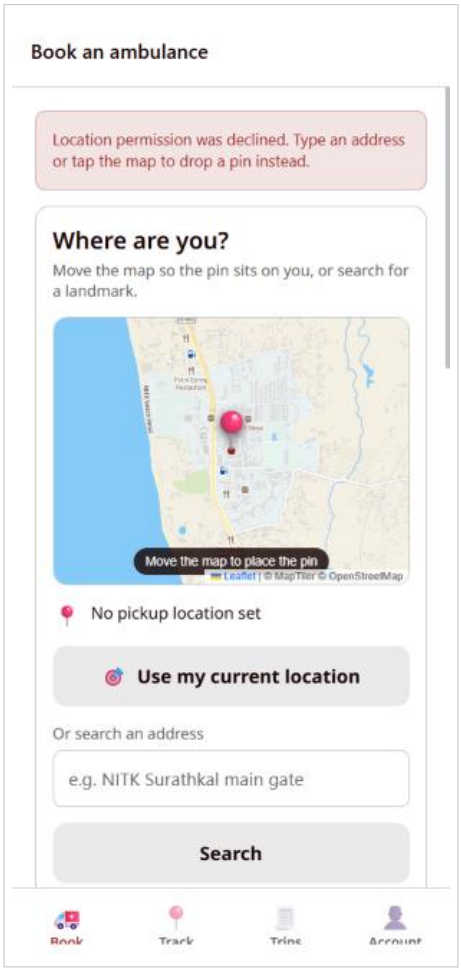


Figure 3. The booking screen. The pin stays at the centre of the map while the map is moved beneath it, and the resolved address is shown below.

3.4 Choosing the class of ambulance

Users need to select the class of ambulance required. The classes available are listed below.

Class	When it should be selected
EMG Emergency	An ambulance is needed immediately and the user does not know which class is appropriate. Every nearby ambulance is alerted at the same moment and the first to accept is assigned.
BLS	Oxygen, a stretcher and a trained attendant.
ALS	Ventilator, defibrillator and a paramedic.
ICU	Critical care with full patient monitoring.
PTV	A planned, non-emergency transfer between facilities.

If in doubt, select Emergency

The four ordinary classes are offered to one ambulance at a time, in order of rank. An emergency request is offered to every eligible ambulance within range simultaneously, and is therefore answered sooner.

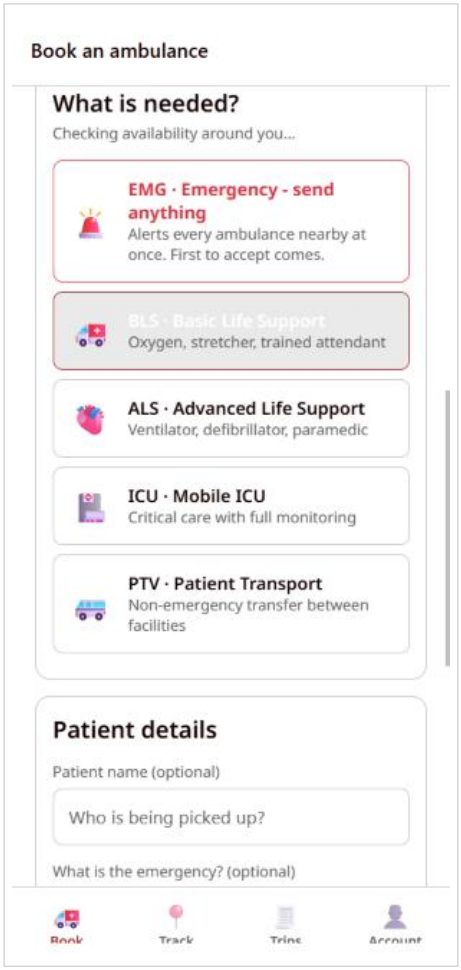


Figure 4. The class selection list. Emergency is placed first because it is the option a user wants when there is no time to choose.

3.5 Placing the request

1. Users need to enter the patient's name and, where possible, a short note describing the emergency. The note is passed to the crew along with the request.
2. The indicative fare is displayed. The final invoice is calculated from the distance actually travelled and will therefore differ from this figure.

3. Click on **Request**.

The application moves to the **Track** tab while an ambulance is sought. Each ambulance is given 45 seconds in which to answer before the request is passed to the next.

3.6 Tracking the ambulance

Once an ambulance has accepted the request, the **Track** tab displays:

- the ambulance moving on a live map, with an estimated time of arrival that is revised as it travels;
- the name of the driver and the registration number of the vehicle;
- the milestones of the trip as they are reached;
- a **Call driver** button, which places a telephone call to the driver. This should be used if the crew are unable to find the patient;
- a **Cancel** button, available until the patient is taken on board.

3.7 Payment, invoice and rating

When the trip is completed, an itemised invoice is raised and appears under **Trips**. The invoice separates the base fare, the distance charge, any class surcharge and tax. Payment may be made by UPI, card, net banking, or in cash on completion.

Users are invited to rate the trip from one to five stars and may add a written comment. The rating contributes to the ranking used when the same ambulance is considered for a future request.

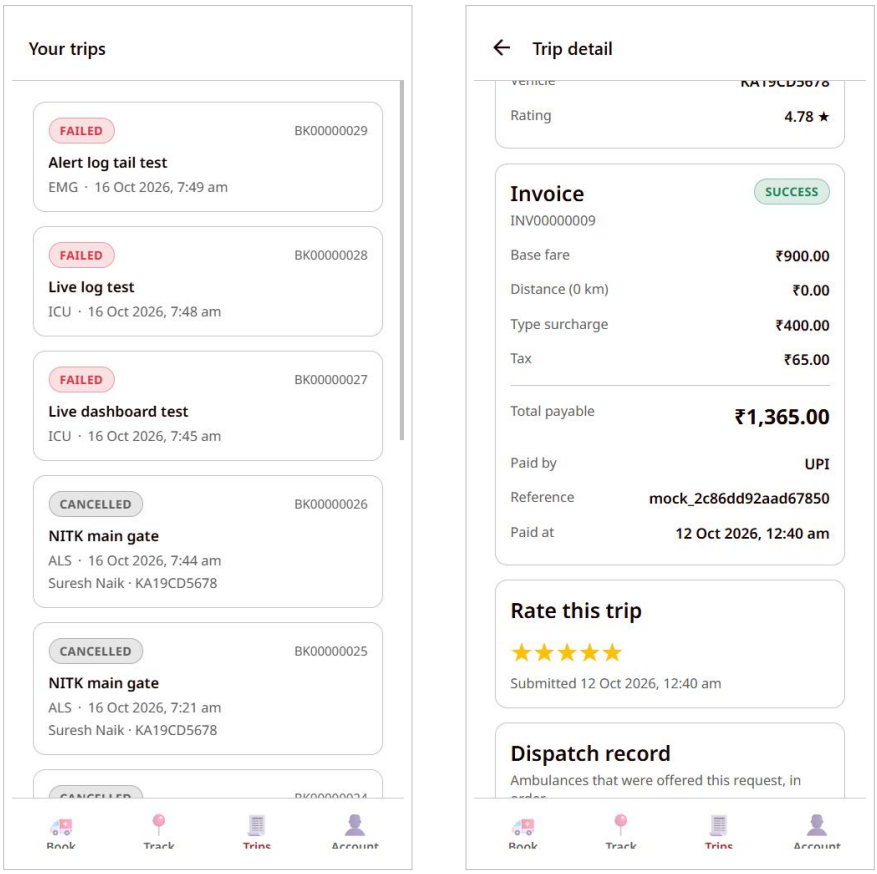


Figure 5. Left: the trip history. Right: the itemised invoice, and the rating given for that trip.

3.8 Reporting an ambulance crew

Where the conduct of a crew has fallen short, the patient may report it. The procedure is described in Section 6, which also explains how such reports are dealt with.

4 The ambulance crew module

Drivers and ambulance operators have access to a certain set of actions that can be performed within the dispatch console. They are illustrated below.

4.1 Actions available to the crew

- Submit a vehicle for verification by the control room.
- Go on duty and off duty.
- Accept or decline a request within the 45 second window allowed.
- Move a trip through its milestones:
  - On the way, Arrived, On board and Complete.
- View the route, the pickup point and the vehicle's own position on a live map.
- View earnings, both settled and outstanding, together with a fortnight of history and every past trip.
- Report a passenger for misconduct on a trip that has ended.

4.2 Submitting the vehicle for verification

A newly registered driver or operator cannot receive any request until the control room has examined their vehicle. Immediately after registration, users are asked to provide its details.

1. Enter the registration number exactly as it is printed on the number plate, without spaces. The format expected is two letters, two digits, two letters and four digits, for example `KA19CD5678`.
2. Select the class of the ambulance: BLS, ALS, ICU or PTV.
3. Enter the driving licence number and, where available, a link to scanned supporting documents.
4. Click on **Submit for verification**.

> **The console remains locked until verification**
> Until an administrator has verified the vehicle, the console stays locked and no requests whatsoever will be received. The console unlocks by itself at the moment verification is granted. Users do not need to sign out and in again, and do not need to keep checking.

4.3 Going on duty

Users need to use the switch at the top of the **Console** tab. While the crew are on duty, the application reports the position of the vehicle approximately every twenty seconds. This is the information by which the vehicle is matched to a nearby patient.

> **Reporting is what keeps a vehicle dispatchable**
> An ambulance whose application has not reported its position for fifteen minutes is taken off duty automatically, because a patient cannot be sent to a vehicle whose position is unknown. Where this occurs the crew are informed, and need only go on duty again.

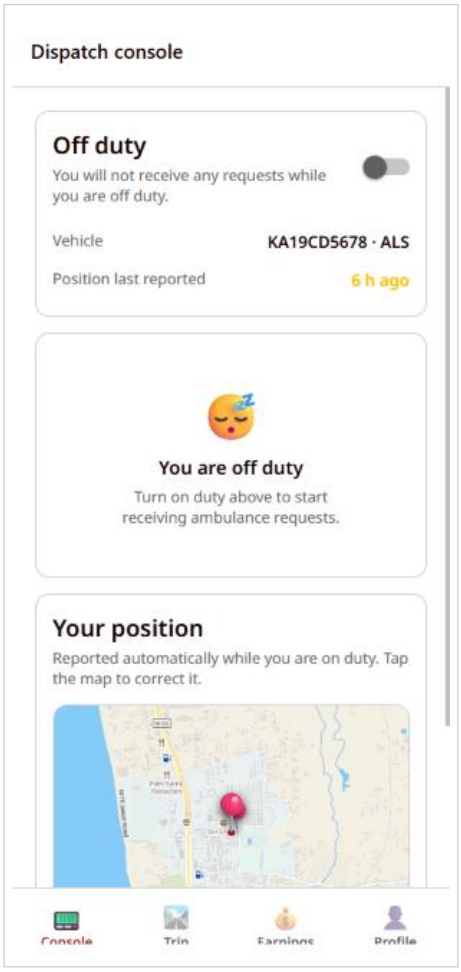


Figure 6. The dispatch console with the crew off duty. The switch at the top of the screen begins and ends the shift.

4.4 Answering a request

A request arrives as a card carrying a countdown, and the device sounds and vibrates. The crew have **45 seconds** in which to answer, after which the request is passed to the next ambulance.

- A card marked **EMERGENCY** was sent to every nearby ambulance at the same moment. The first crew to accept are assigned, so hesitation costs the job rather than delaying it.
- Declining passes the request on immediately. A decline is not held against the crew.

> **Alerts require the application to be running**
> The application may be in the background and alerts will still sound. If it has been closed from the list of recent applications, however, nothing can reach it. Crews need to leave the application running for the duration of their shift.

4.5 Running the trip

The **Trip** tab shows the pickup point, the route and the vehicle's own position, and advances through the milestones one at a time. A milestone cannot be skipped.

Milestone	To be marked when
On the way	The crew set off towards the pickup point.
Arrived	The crew reach the pickup point.
On board	The patient is inside the ambulance. **The distance charge begins here.**
Complete	The patient has been handed over. The invoice is raised.

> **The fare is measured between two of these taps**
> The distance billed is measured by GPS between **On board** and **Complete**, and not from any figure entered by hand. Marking **On board** late, or **Complete** early, reduces the amount billed.

4.6 Earnings

The **Earnings** tab totals the amounts settled and outstanding, and lists a fortnight of history together with every past trip. A trip in this list may also be the subject of a report, as described in Section 6.

5 The control room module

Administrators have access to a certain set of actions that can be performed within the control room. They are illustrated below. Every screen in this module updates by itself as events occur; none of them needs to be reloaded.

5.1 Actions available to the administrator

- Monitor the operational position of the whole service in real time.
- Verify, reject or suspend an ambulance submitted by a driver or operator.
- Search, filter and inspect every booking, and for any one of them:
  - re-run matching on a booking that failed to find an ambulance;
  - force the cancellation of a booking that has become stuck.
- Follow the entire fleet on a map, filtered by availability.
- Review reports made by crews against passengers and by patients against crews, and decide the outcome.
- Produce reports on response time, demand, revenue and provider performance over a chosen date range.
- Inspect the log of every alert the system has sent.
- Inspect the audit trail of every decision and action taken.
- Suspend or reinstate any account.

5.2 The dispatch overview

The control room opens on the dispatch overview, which carries the live operational picture.

Figure	Meaning
Awaiting a match	Requests that are still being offered to ambulances.
Trips in progress	Trips from assignment through to the patient being on board.
Ambulances free	Vehicles that are verified, on duty and reporting their position: that is, the vehicles that could actually be dispatched at this moment.
Not reporting	Vehicles that are on duty but whose application has fallen silent. These require attention, as they cannot be dispatched.

A panel appears at the head of the screen whenever a provider account is waiting to be verified, since an unverified ambulance receives no requests at all.

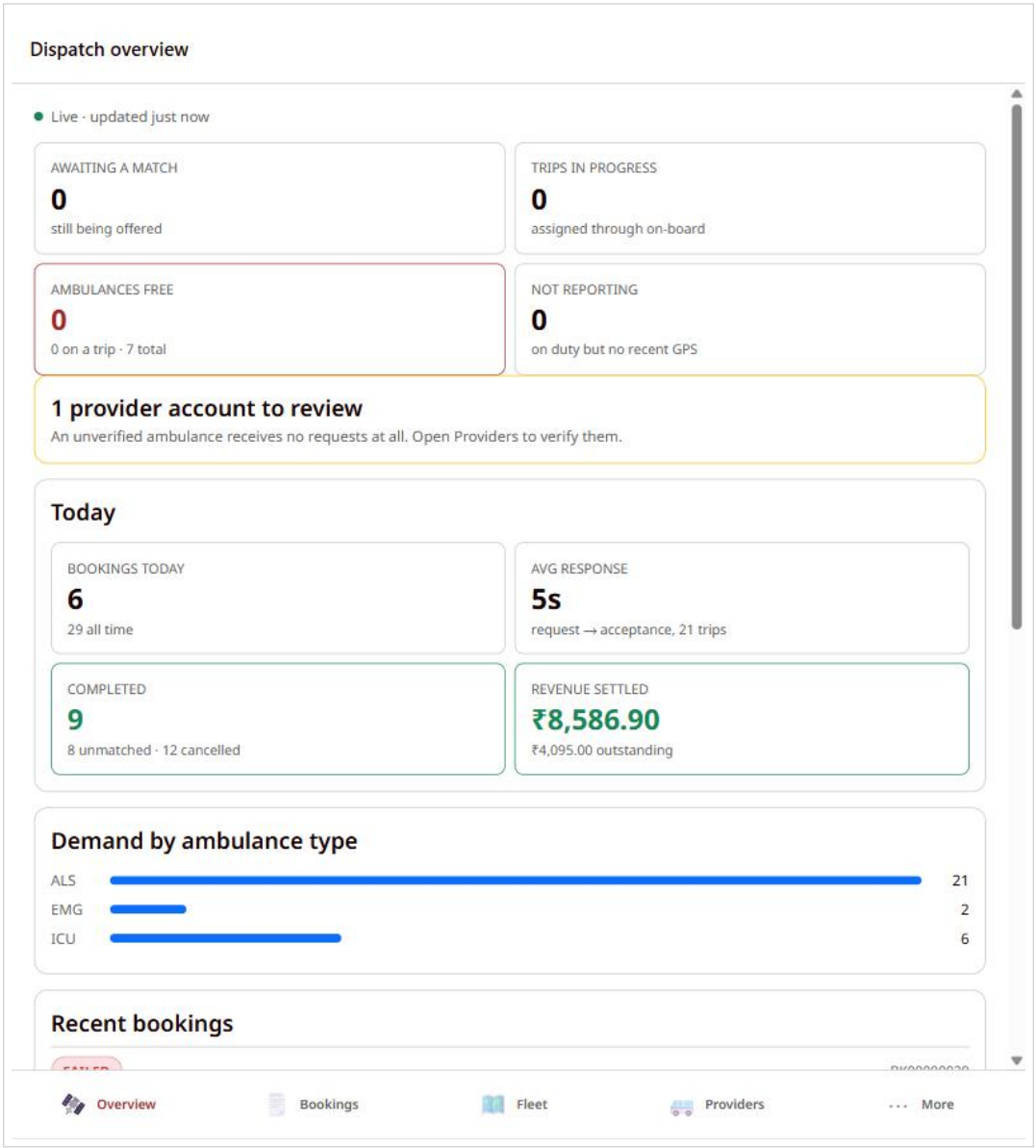


Figure 7. The dispatch overview, shown in a desktop browser. The figures and the charts below them are revised as events occur.

5.3 Verifying an ambulance

1. Open the **Providers** tab and filter by **PENDING**.

2. Open the account under examination and read the registration number, the class of vehicle, the licence number and any documents supplied.
3. Click on **Verify** or on **Reject**, adding a remark that records the reason for the decision.

The driver is informed at once and their console unlocks by itself. This step is what places a vehicle into service, and until it is taken the vehicle receives nothing.

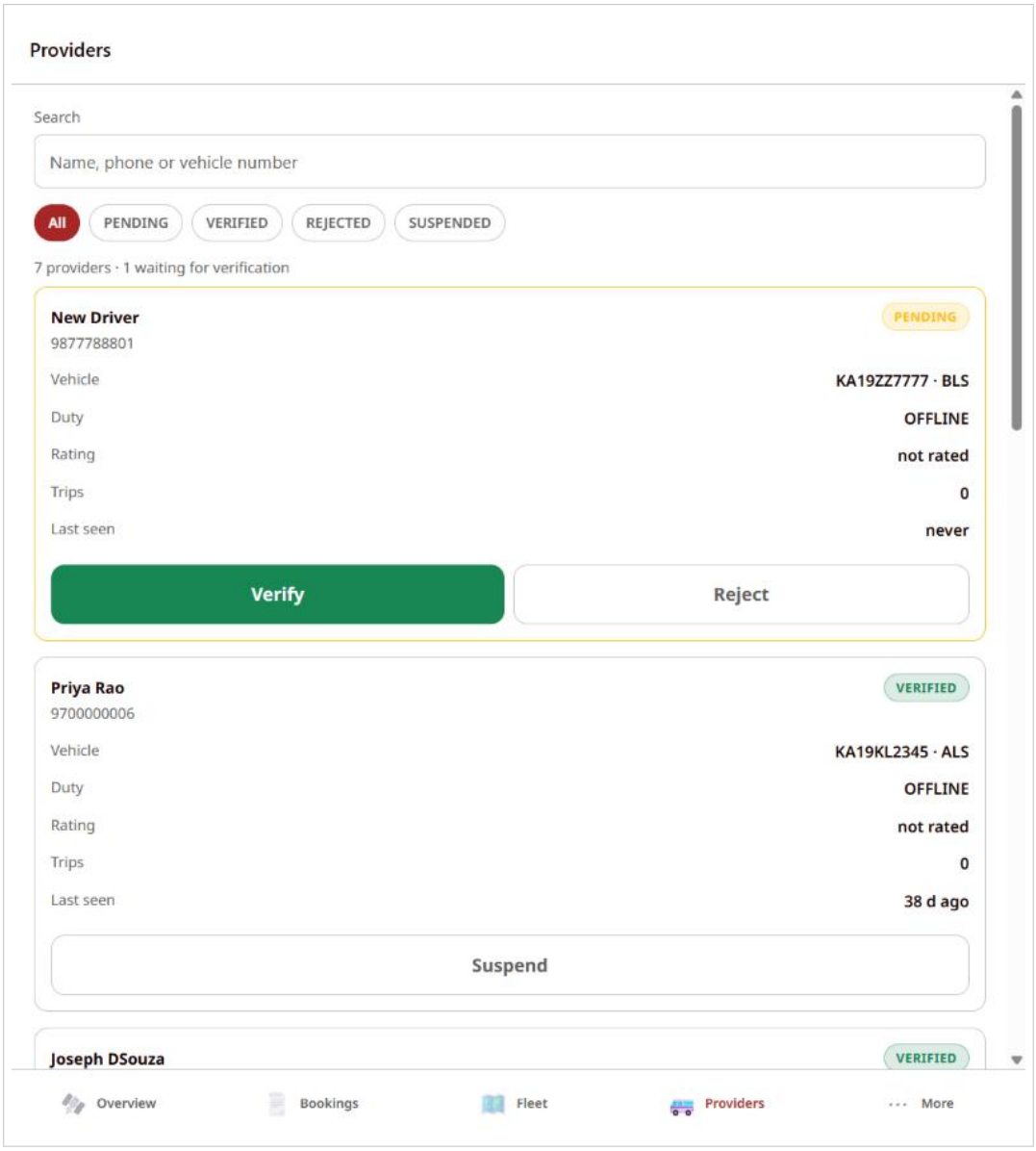


Figure 8. The provider list. An account awaiting verification is highlighted and carries the **Verify** and **Reject** actions.

5.4 Monitoring bookings

The **Bookings** tab lists every booking in the system and may be filtered by status, by class of ambulance or by date, or searched by booking identifier, address, patient name or telephone number.

Opening a booking presents the complete case file: who requested it, every ambulance to which it was offered and what each of them answered, the GPS trace of the journey, the invoice, the rating and every alert sent in connection with it. Two corrective actions are

available from this screen:

- **Re-dispatch** runs the matching procedure again on a booking that failed to find an ambulance;
- **Force cancel** closes a booking that has become stuck.

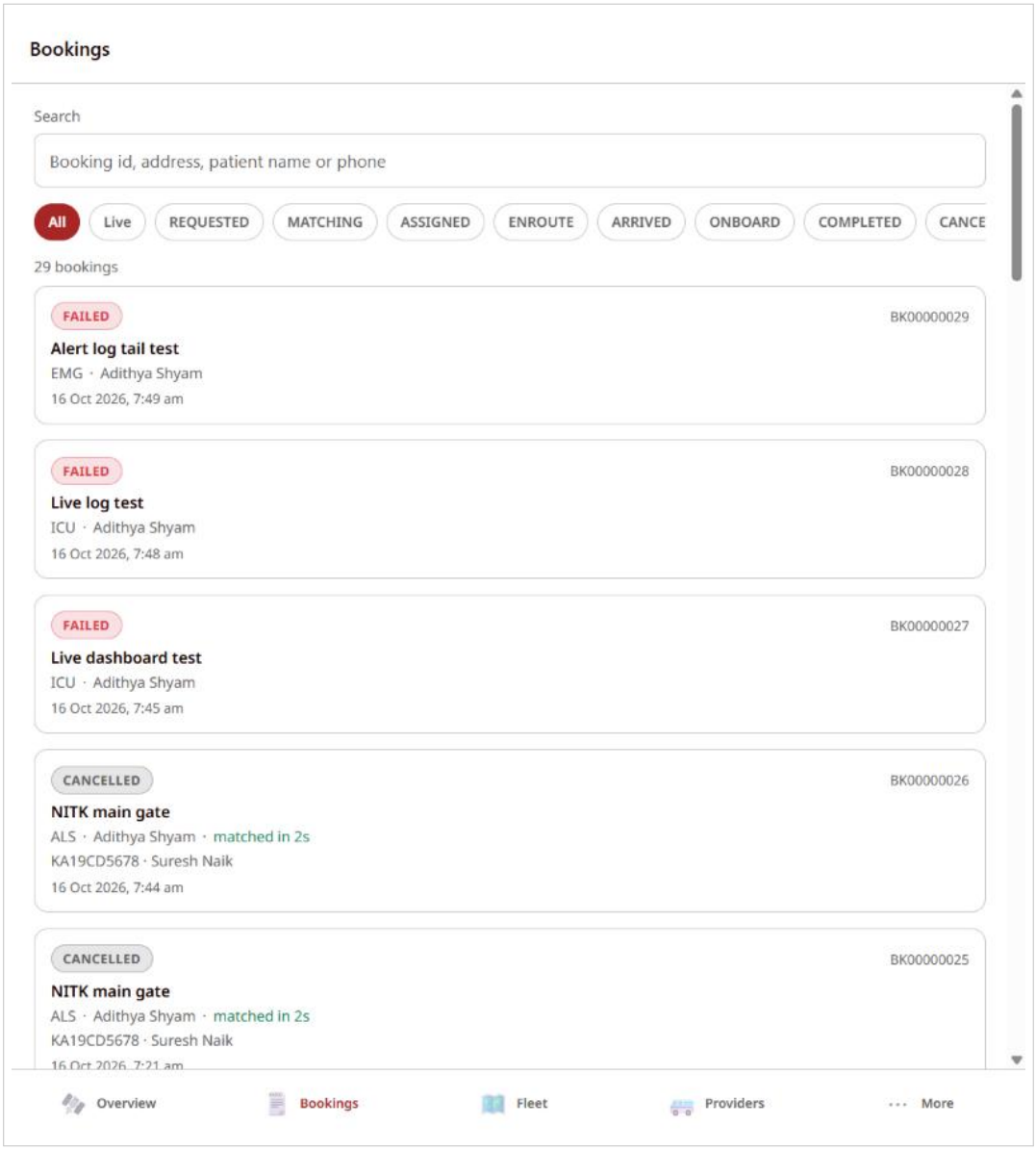


Figure 9. The booking list, filtered by status. The identifier of each booking is shown on the right.

5.5 The fleet map

The **Fleet** tab plots every vehicle known to the system on a map, colour coded by availability, with the time at which each last reported its position. The list beneath the map may be filtered to vehicles that are available, busy or offline.

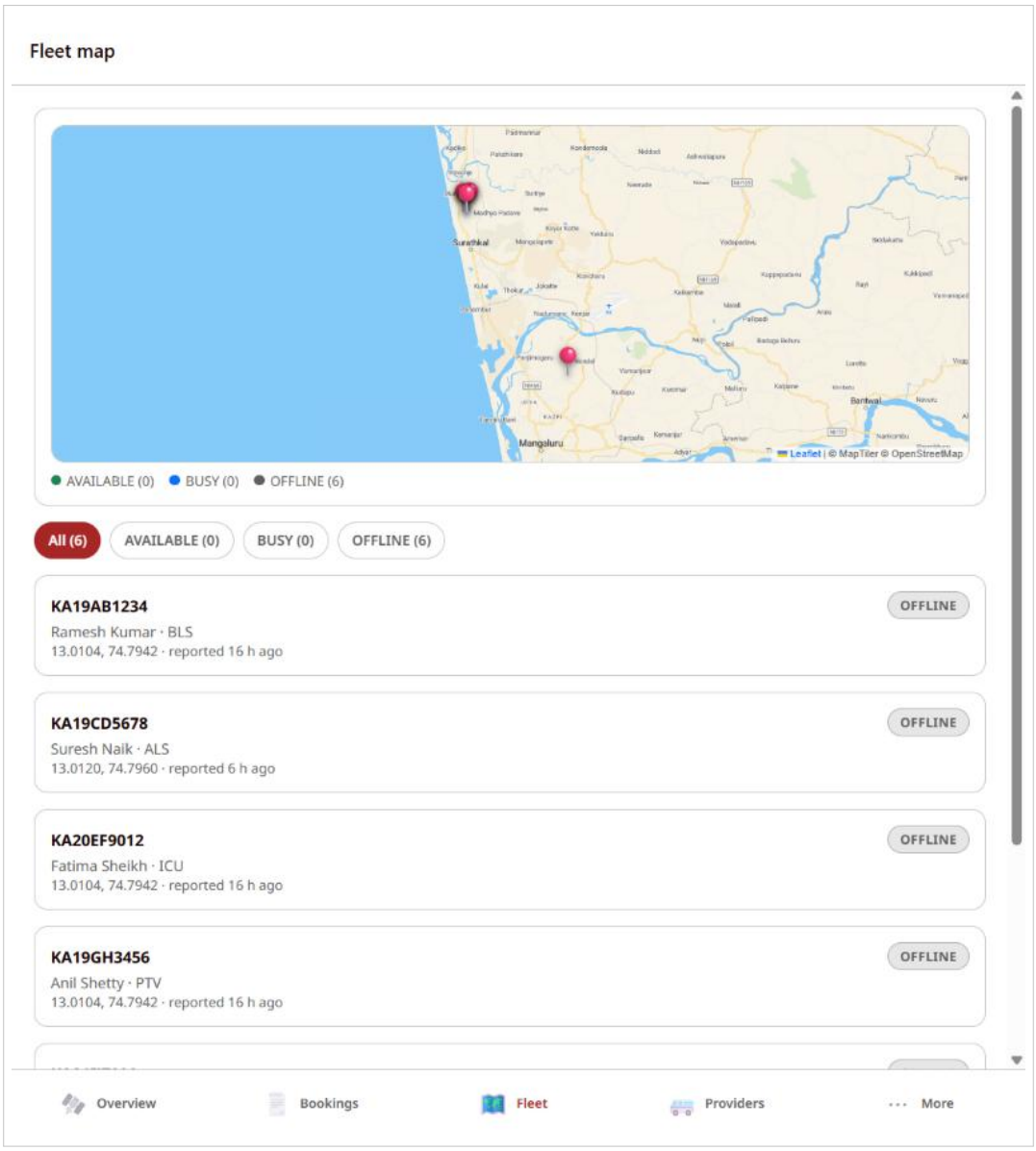


Figure 10. The fleet map. The time since each vehicle last reported is the figure that determines whether it can be dispatched.

5.6 Reports, logs and accounts

The remaining functions of the control room are reached through the **More** menu.

Entry	Contents
Passenger reports	Complaints made by crews against passengers and by patients against crews, with the actions available against each. See Section 6.
Reports	Response time, demand by class, revenue and provider performance over a chosen date range.
Alert log	Every WhatsApp message, email and push notification the system has sent, with its delivery state.
Audit trail	Every decision and action taken in the system, in order, including those of the administrator reading it.
Accounts	Every patient, driver and operator, with the facility to suspend or reinstate an account.

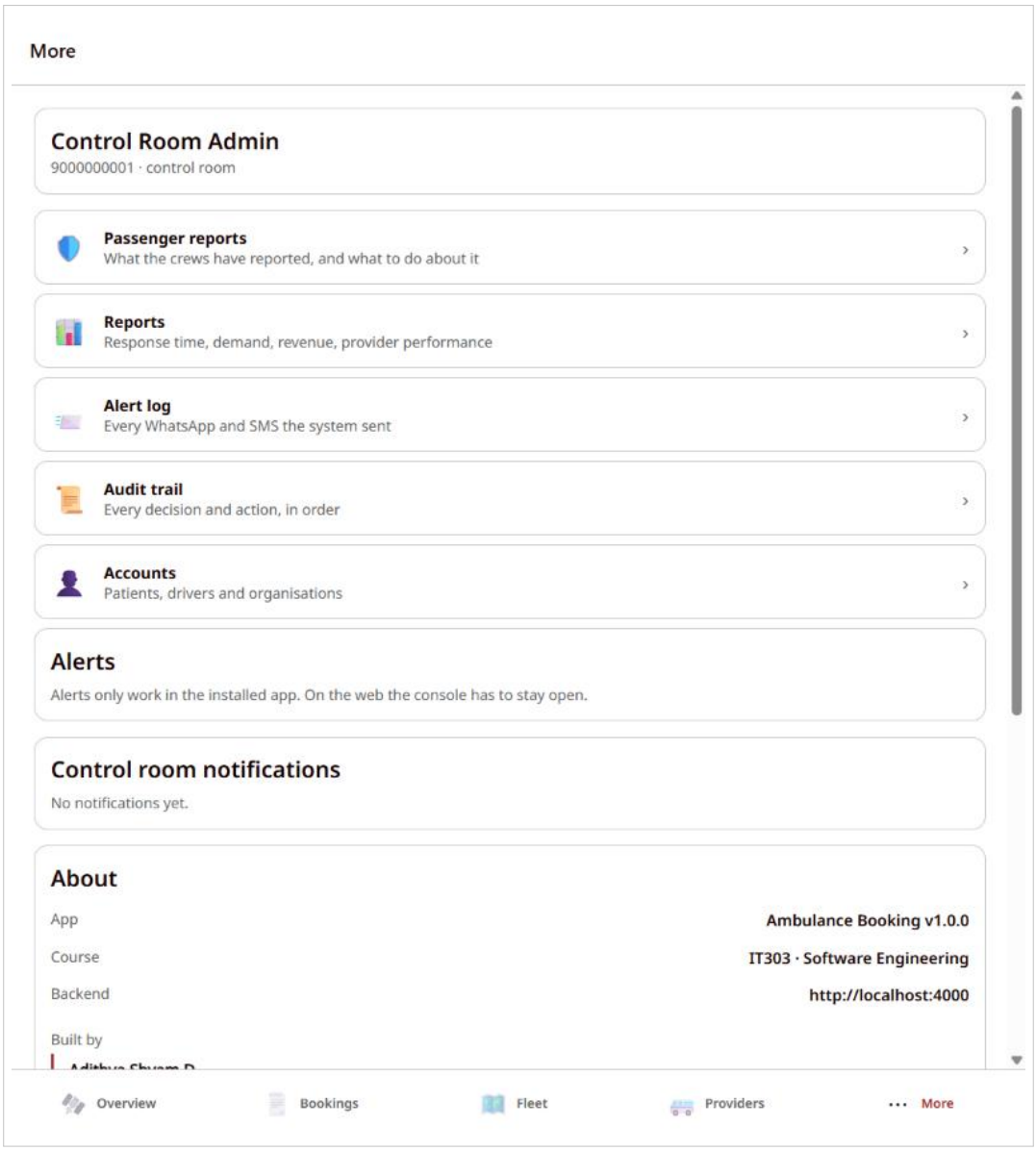


Figure 11. The control room menu, from which the reporting, logging and account functions are reached.

6 Reporting misconduct

A hoax request occupies an ambulance that another patient needed, and a crew who behave badly should be answerable for it. The system therefore allows a report to be made in either direction, and the procedure is deliberately the same both ways.

> **Making a report decides nothing by itself**
> Filing a report has no effect whatsoever on the account of the person reported. The report is placed before the control room, who decide the outcome. No account is suspended merely because somebody has said that it should be.

6.1 A crew reporting a passenger

1. Open the **Earnings** tab and locate the trip concerned.
2. Click on **Report this passenger**.
3. Select what occurred: a hoax call, the passenger not turning up, abusive or threatening behaviour, a refusal to pay, or some other matter.
4. Describe what occurred, briefly and factually, and submit the report.

6.2 A patient reporting a crew

1. Open the **Trips** tab and locate the trip concerned.
2. Click on **Report this ambulance**.
3. Select what occurred: the ambulance never arriving, unsafe driving, rude or abusive behaviour, a demand for more than the invoiced fare, or the condition of the vehicle or its equipment.
4. Describe what occurred and submit the report.

In both directions, a report may only be made by a party to the trip, only once per trip, and only after the trip has ended.

6.3 How the control room decides

Reports are listed under **Passenger reports** in the **More** menu. Each entry shows the person reported, the person who reported them, what was said, and **the number of other reports naming the same person together with the number that were upheld**. One complaint is an incident; four are a pattern, and that distinction is the judgement the administrator is being asked to make.

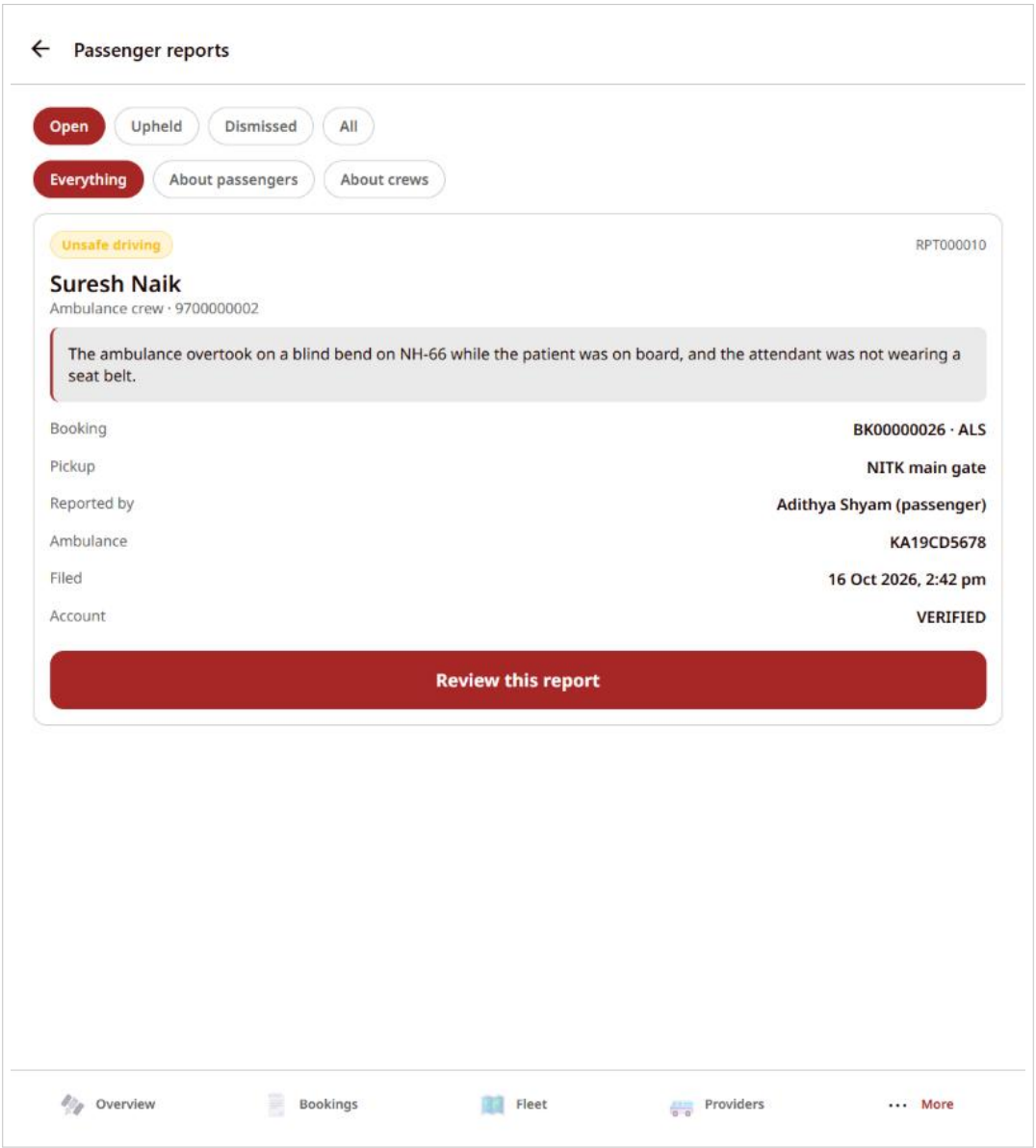


Figure 12. A report awaiting review. The booking, the pickup point, the vehicle and the account status of the person reported are all shown alongside the complaint itself.

Decision	Effect
Dismiss	No action is taken. A report found to be baseless costs the person reported nothing, and so no penalty may be attached to a dismissal.
Uphold — warn	Recorded against the account, with no further effect.
Uphold — suspend	The account is signed out everywhere and locked. A suspended driver is additionally taken off duty, so that no further request can reach them. The suspension may be reversed from **Accounts**.
Uphold — delete	The name, telephone number and email address are erased and the account is locked. The bookings, invoices and the report itself are retained, because the record has to outlive the person it concerns. This cannot be reversed.

Whoever made the report is informed of the outcome in either case.

7 Troubleshooting

Message or symptom	Action to be taken
The server did not answer within a minute	The hosted backend is put to sleep when it is not in use and takes roughly fifty seconds to wake. Users need to try the operation again; the second attempt is prompt.
Cannot reach the server	Confirm that the device is online. Where a backend of one's own is being used, confirm the address shown on the sign-in screen under **Backend …Change**.
No GPS fix yet	Move to a position with a clearer view of the sky, or set the pickup point on the map instead. The application does not wait indefinitely for a fix.
The verification code never arrives	Confirm the spelling of the email address and examine the spam folder. An address at a domain that does not exist can never be delivered to.
No ambulance could be assigned	No vehicle was free within fifteen kilometres. Users need to try again, select **Emergency** so that every nearby ambulance is asked at once, or telephone 108.
The dispatch console remains locked	The vehicle has not yet been verified by the control room. The console unlocks by itself; nothing needs to be restarted.
No alert is received when a request arrives	Open **Profile**, then **Alerts**, and click on **Send a test alert**. Alerts require the application to be running; the background is sufficient, a closed application is not.
The account is suspended	An administrator has suspended the account following a report. Users need to contact the control room.

A Demonstration accounts

The database supplied with the system is seeded with the accounts below, so that each role may be examined without registering.

Role	Mobile number	Password
Patient	9800000001	`user123`
Driver (ALS)	9700000002	`driver123`
Driver (ICU)	9700000003	`driver123`
Control room	9000000001	`admin123`

These are demonstration credentials

The accounts above exist only in the seeded development database and carry no real personal data. They should be removed before the system is put to any use beyond assessment.

B Requirement traceability

The table below maps each functional requirement of the Software Requirements Specification onto the section of this manual in which it is demonstrated.

SRS	Requirement	Demonstrated in
F.1	Registration, OTP verification and password recovery	Sections 2.2 to 2.4
F.2	Role-based login	Section 2.1, Section 2.3
F.3	Select ambulance type and place booking	Sections 3.4 and 3.5
F.4	Capture pickup location	Section 3.3
F.5	Match nearest available ambulance	Sections 3.4 and 5.2
F.6	Notify driver of a request	Section 4.4
F.7	Accept or reject a request	Section 4.4
F.8	Live tracking and estimated time of arrival	Sections 3.6 and 5.5
F.9	Trip status updates	Section 4.5
F.10	Payment and billing	Section 3.7
F.11	Booking history and records	Sections 3.7 and 4.6
F.12	Rating and feedback	Section 3.7
F.13	Administrative management, reports and sanctions	Sections 5 and 6
F.14	Notifications and alerts	Section 4.4, Section 5.6, Section 7

A fuller matrix, which continues from each requirement into the module and the source file that implements it, is maintained in `docs/TRACEABILITY.md` in the project repository.

C Document control

Version	Date	Summary of change
1.0	16 October 2026	First issue, covering the patient, crew and control room modules.

Ambulance Booking System — User Manual, version 1.0

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